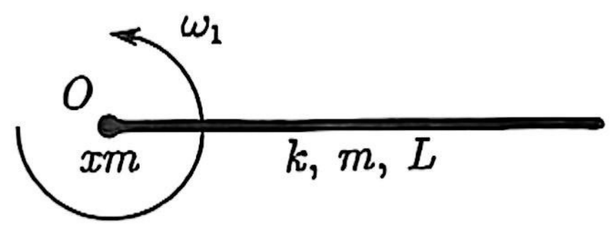
1. (40 分)



解: (1) 位于 $r\sim r+dr$ 段

$dT = \frac{kL}{dx}(dr-dx) = kL\left(\frac{dr}{dx}-1\right)$

$\therefore\ dT = kL\frac{d^2r}{dx^2}dx$

$\therefore\ kL\frac{d^2r}{dx^2}dx = -\frac{m}{L}dx\,\omega_1^2 r$

即 $\frac{d^2r}{dx^2} = -\frac{\pi^2}{9}\frac{r}{L^2}$

$\therefore\ \frac{dr'}{dr}r' = -\frac{\pi^2}{9}\frac{r}{L^2} \Rightarrow r'^2 = -\frac{\pi^2}{9}\frac{r^2}{L^2}+C_1$

移项并积分:

$\frac{3L}{\pi}\arcsinh\sqrt{\frac{\pi^2}{9C_1L^2}}\,r = x+C_2$

$\therefore\ r = \sqrt{\frac{9L^2C_1}{\pi^2}}\sinh\frac{\pi x}{3L}$

$\therefore\ \sqrt{C_1}\cosh\frac{\pi x}{3} = 1 \Rightarrow \sqrt{C_1} = \frac{1}{\cosh\frac{\pi}{3}}$

$\therefore\ r = \cancel{\frac{3L}{\pi\cos\frac{\pi}{3}}\sinh\frac{\pi x}{3L}}\ \frac{3L}{\pi}\frac{\sinh\frac{\pi x}{3L}}{\cos\frac{\pi}{3}}$

$R = \frac{3L}{\pi}\tan\frac{\pi}{3} = 0.\cancel{7461L}\ 1.65L$

(2) $I_1 = \int\frac{m}{L}dx\,r^2 = \int_0^L\frac{9}{\pi^2}mL\frac{\sinh^2\frac{\pi x}{3L}}{\cos^2\frac{\pi}{3}}dx$

$\cancel{= \frac{9}{\pi^2}mL\Big/\cosh^2\frac{\pi}{3}\cdot\left(\frac{1}{4}\sinh\frac{2\pi}{3}-\frac{L}{2}\right)}$

$\therefore I_1 = mL^2\cdot\frac{9}{\pi^2\cos^2\frac{\pi}{3}}\left(-\frac{3}{4\pi}\sinh\frac{2\pi}{3}+\frac{1}{2}\right)$

$I_1 = \cancel{0.162mL^2}\ 1.07mL^2$

(3) 此时 $r = \sqrt{\frac{36L^2}{\pi^2}C_2}\sinh\frac{\pi x}{6L}$

$\therefore\ kL\left(\sqrt{C_2}\cos\frac{\pi}{6}-1\right) = xm\omega^2\frac{6L}{\pi}\sqrt{C_2}\sin\frac{\pi}{6}$

① ② $I = xm\,C_2\sinh^2\frac{\pi}{6}\cdot\frac{36L^2}{\pi^2} + \int_0^L\frac{m}{L}dx\cdot\frac{36L^2}{\pi^2}C_2\sinh^2\frac{\pi x}{6L}dx$

③ ④ $I_1\omega_1 = I\omega_2$

⑤ 得 $x = 0.726$.

⑦ ⑧ ⑨ ⑩ ⑪ ⑫ ⑬ ⑭ ⑮ ⑯